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ML, DL, NLP Boot camp
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Sec-20

lec-1

What is

Statistics

its Applications

lec-2 Types of Statistics

Statistics is a field that deals with the collection, organization, analysis & interpretation of data



Basically it helps in Decision Making.

Ex: Age = { 24, 27, 10, 13, 28, 29, 31, 32 }



Online Shopping.

(*)

Now an important thing to segregate the online shoppers and hence we will have the marketing strategies for promotion & other

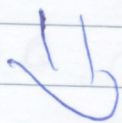
Descriptive statistics, and

- (1) Mean
- (2) Median
- (3) Distribution of Age

→ It can be used to
analyse & summarise
statistical data

Charts & Graphs

- (1) Histogram
- (2) Probability Distribution
Function
- (3) Cumulative Density Function



We understand the data
→ make decisions for
your business.

"Let's understand a simple Example:-"

Example: There is a Bank :-

At Location A, there is an ATM

Location A
ATM

"Now they move the branch in Location B"

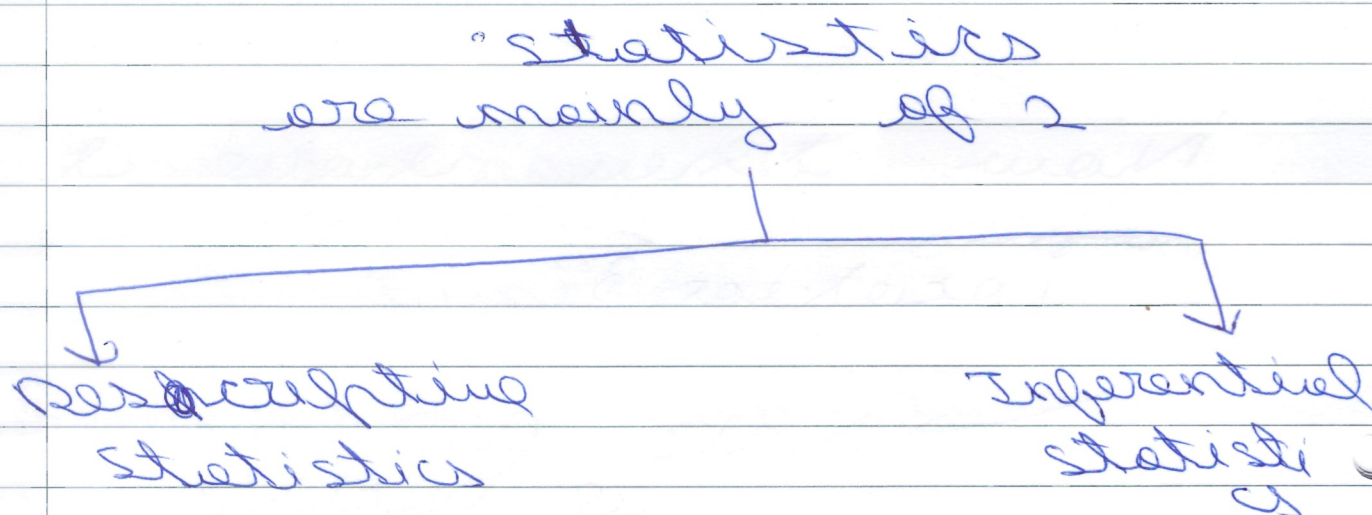
→ Now for making a decision,

they will have to calculate based on the data of location A, ~~that~~ but they will accordingly make decision.

Zero Applications are:-

- (1) Machine Learning
- (2) DATA Analytics
- (3) Business Analytics
- (4) Risk Analytics

Now this was a basic analysis of statistics.



Descriptive Statistics

It consists of organising & summarizing the data.

Example:- Measures of central tendency like Mean, Median & Mode.

(5)

The other method of
measure of
dispersion like-

- (i) Variance
 - (ii) Standard Deviation
-

⇒ Now coming to
Inferential Statistics.

It means the statistician
to draw some
conclusions or inferences
using some experiment.

Some examples of
experiments are like

- (1) Z-test
- (2) t-test

We collect the data
to collect as sample data.

By experiments, we
calculate the conclusions
which are made for
some other data.

There is no need to collect data for other things.

→ This is called as population data.

Now we can think of an example:-

→ There is a college with 1000 students.

There is a class of statistics students.

{ 180 cm, 170 cm, 165 cm, 150 cm,
160 cm }
→ Height of students

⑧ Now, I need to make inferences from data.

Like in Descriptive Statistics, we can

Mean, median, ~~Average~~ = 165
Average = 165

⑦

Now coming to Inference:-

Inference:- We use many
conclusions &
inferences.

Like we ~~not~~ perform
experiment on the data
which is then used to
the 1000 students height
conclusion.